Lab 2: Exploration by visualization: the streaming movies dataset

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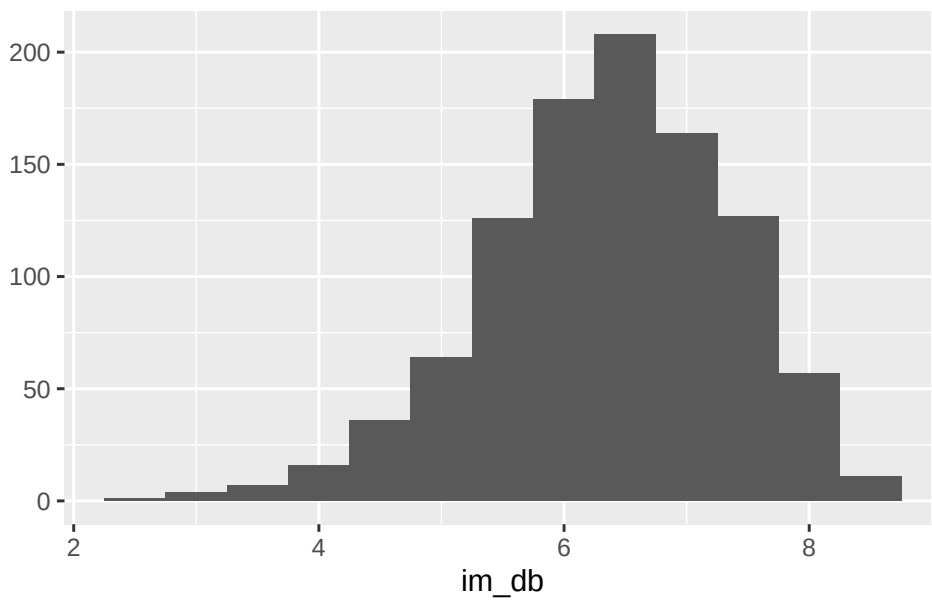
Visualization by example

Exercise 1

```
qplot(x = im_db, binwidth = 0.5, data = streaming)
```

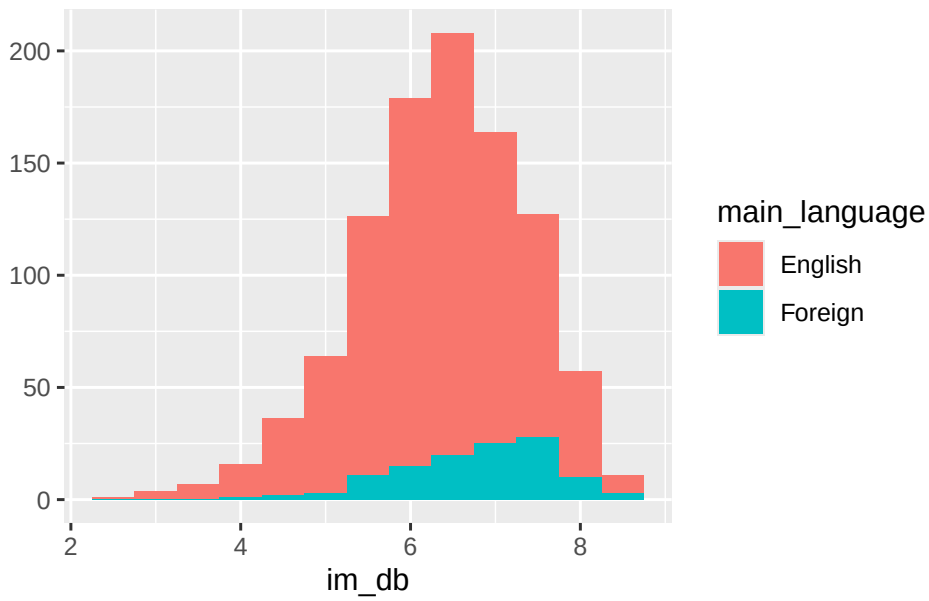
```
## Warning: `qplot()` was deprecated in ggplot2 3.4.0.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



i. The histogram organized the distribution of IMDB ratings, listing them on the x-axis. The bin width indicates that the range of values that each bin covers is 0.5.

Exercise 2

```
qplot(
  x = im_db,
  binwidth = 0.5,
  fill = main_language,
  data = streaming
  )
```



i. The fill argument color-codes languages based on whether they are English or Foreign and depicts this within the histogram.

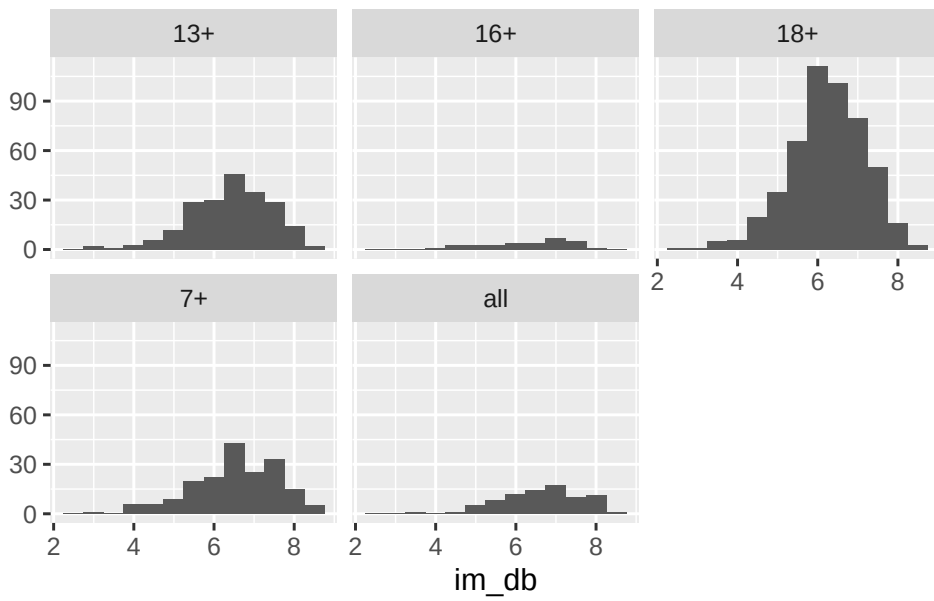
ii. The language of most of the movies in this database is English.

Exercise 3

i. The shape of the English distribution appears to be left-skewed and unimodal. The ratings seem to be centered around a rating of 6. The Foreign IMDB ratings' distribution appears to be left-skewed and could be considered unimodal too, but it appears centered around a rating of 7.5, and it is significantly flatter than the English distribution. The distribution also does not cover as much area. Because of this, I can conclude that there appears to be a tangible difference in the average IMDB rating for these two distributions. Based on my experience of watching movies in recent years, this a result that I would have expected to see. For studios like Marvel and Disney, there seems to be a decline in creativity and originality, which may be disliked by audiences. There are some great films that have been released. They are, however, likely outnumbered by mediocre ones.

Exercise 4

```
qplot(
  x = im_db,
  binwidth = 0.5,
  facets = ~ age,
  data = streaming
  )
```



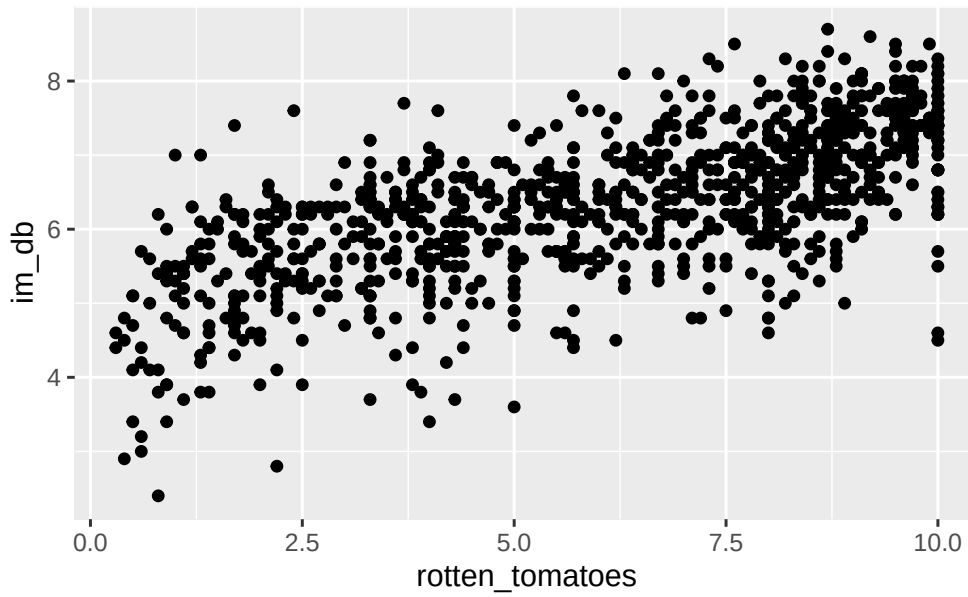
i. There are 4 facets, excluding "all".

ii. Each faceted sup-plot represents a different age group.

iii. The 18+ facet's distribution contains the most movies.

Exercise 5

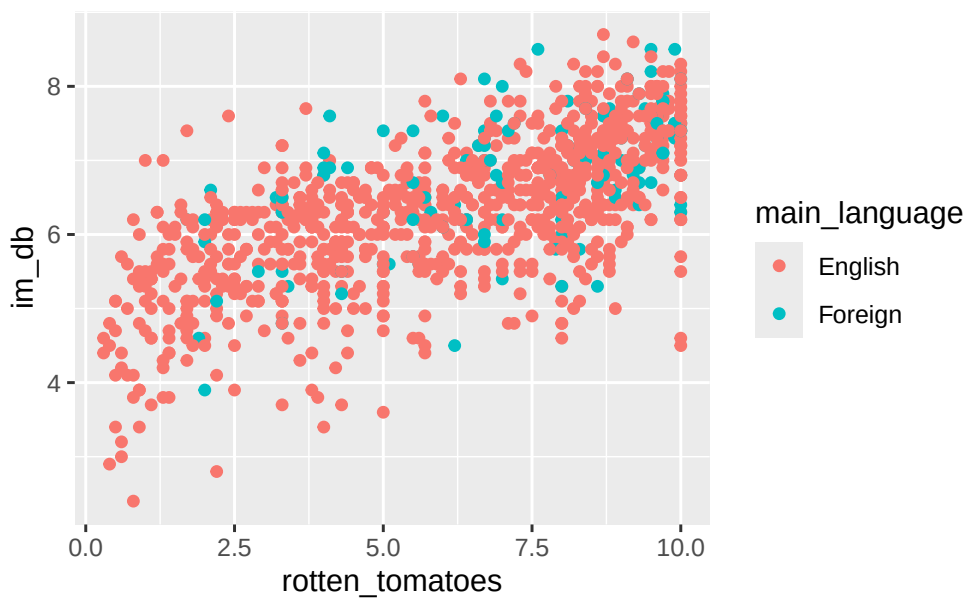
```
qplot(x = rotten_tomatoes, y = im_db, data = streaming)
```



i. The density of points increases with ratings above 7.5. The scatterplot appears to have a positive linear association.

Exercise 6

```
qplot(x = rotten_tomatoes, y = im_db, color = main_language, data = streaming)
```

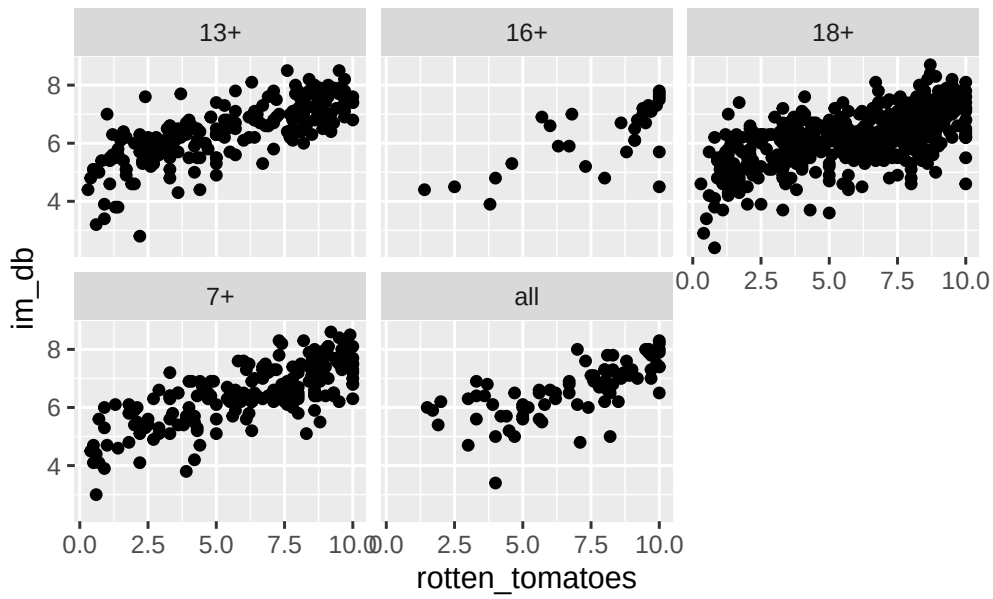


i. The overall trend for different languages is similar between ratings on IMDB and Rotten Tomatoes, but English movies tend to receive higher ratings than foreign language movies.

Ex. There are many more English movies that received ratings of 10.0 than foreign language movies.

Exercise 7

```
qplot(x = rotten_tomatoes, y = im_db, facets = ~ age, data = streaming)
```

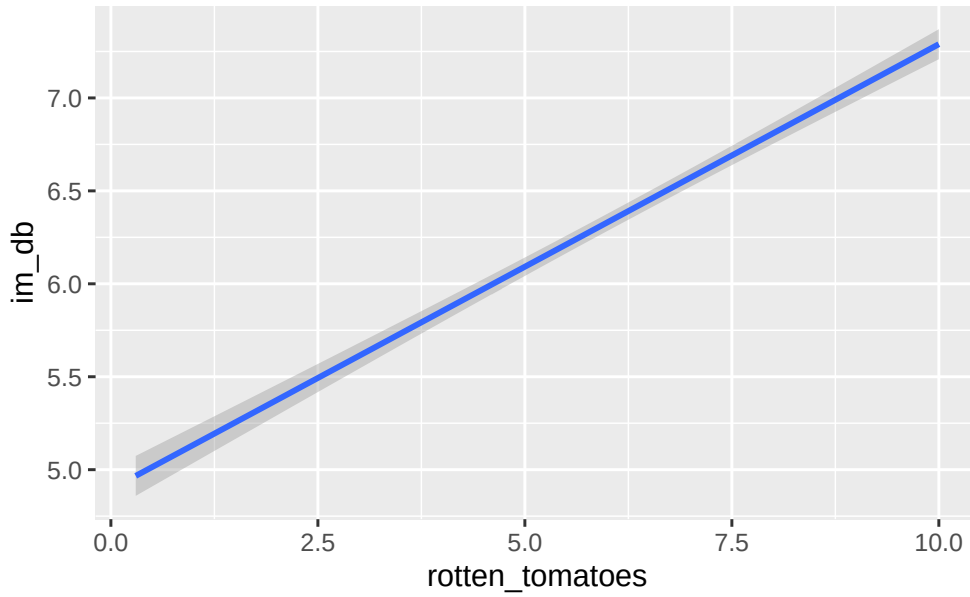


i. The data is the same, but the insights we can pull from the information are different because we are able to discern trends from each age group. While some facets are denser with information than others, all of them appear to have a positive linear relationship.

Exercise 8

```
qplot(
  x = rotten_tomatoes,
  y = im_db,
  geom = "smooth",
  method = "lm",
  data = streaming
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



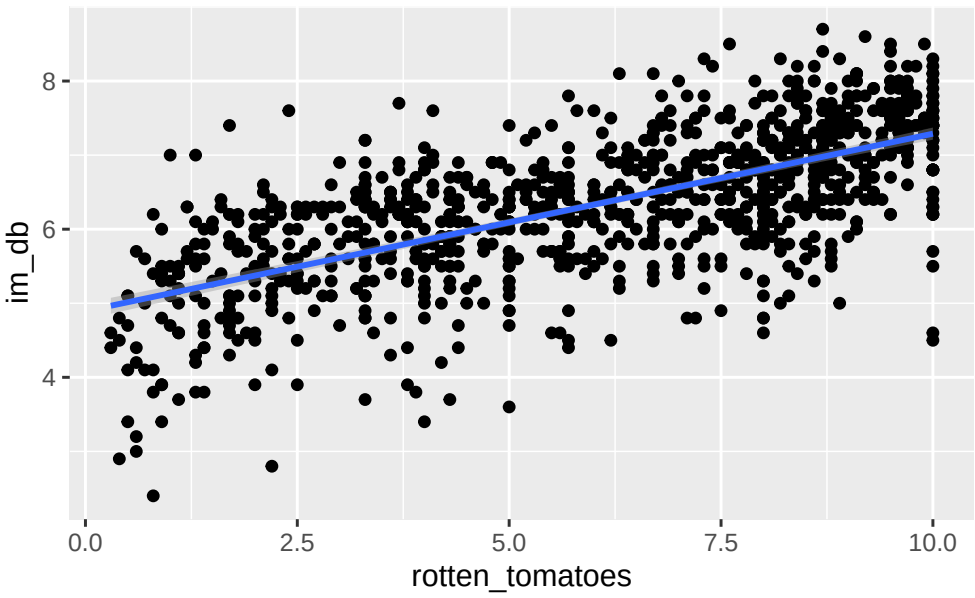
i. This graph aligns with my assertion that the relationship between IMDB and Rotten Tomatoes ratings is positive and linear.

Exercise 9

```
qplot(
  x = rotten_tomatoes,
  y = im_db,
  geom = c("point", "smooth"),
  method = "lm",
  data = streaming
  )
```

```
## Warning in geom_point(method = "lm"): Ignoring unknown parameters: `method`
```

```
## `geom_smooth()` using formula = 'y ~ x'
```


8
6
4
im_db
0.0
2.5
5.0
7.5
10.0
rotten_tomatoes